Question 1: Model it

The ontology has **seven node types and 17 relationship types**, comprising eight inverse pairs and one symmetric relationship. It covers ten companies in the semiconductor supply chain: AMD, Applied Materials, ASML, Broadcom, Intel, KLA, Lam Research, Micron, NVIDIA and TSMC.

The aim is to build a small, connected model that supports questions about competition, supplier concentration and indirect dependencies. Companies offering the same product may compete, while several companies depending on one supplier may indicate concentration risk. These connections identify areas for investigation; they do not independently establish a bottleneck.

I used Protégé and its reasoner to check the ontology's logical consistency. This helps identify contradictions in the declared rules, but does not prove that the model is complete or that extracted facts are correct.

Node types

- **`Company`** - A business in the ten-company basket. Supplier and customer are roles within relationships; a company can be both.
- **`Product`** - A physical good or identifiable product family, such as a memory chip or lithography system.
- **`Service`** - A commercially offered activity, such as wafer fabrication, semiconductor packaging or testing.
- **`SourceDocument`** - A specific document supporting an extracted fact, such as a regulatory filing or earnings release.
- **`Topic`** - A theme relevant to a company, product, service, document or event, such as AI demand or export controls.
- **`Sector`** - An industry category used to classify a company's business activities.
- **`Event`** - A specific occurrence at an identifiable time, such as an earnings announcement, guidance revision or supply disruption.

The classes are mutually disjoint, except that `Topic` and `Sector` may overlap: an industry category can also be a subject of discussion.

Relationships

- **`supplies_to`: Company → Company** - Connects a supplier to a direct customer. A multi-step supply path does not imply a direct supply relationship.
- **`buys_from`: Company → Company** - Connects a customer to a direct supplier. Inverse of `supplies_to`.
- **`competes_with`: Company ↔ Company** - Connects companies that compete in at least one product or service market. Competition is symmetric, but does not propagate through chains.
- **`offers`: Company → Product** - Identifies a product a company offers commercially. This does not necessarily mean the company manufactures it.
- **`offered_by`: Product → Company** - Identifies a company offering the product. Inverse of `offers`.
- **`provides`: Company → Service** - Identifies a service a company provides commercially.
- **`provided_by`: Service → Company** - Identifies a company providing the service. Inverse of `provides`.

- **`operates_in`: Company → Sector** - Classifies a company's business activities. A company may operate in multiple sectors.
- **`includes`: Sector → Company** - Identifies companies operating in a sector. Inverse of `operates_in`.
- **`has_topic`: Company / Product / Service / SourceDocument / Event → Topic** - Associates an entity with a relevant theme. Sharing a topic does not establish a commercial relationship.
- **`topic_of`: Topic → Company / Product / Service / SourceDocument / Event** - Identifies entities associated with a theme. Inverse of `has_topic`.
- **`concerns`: Event → Company** - Identifies a company directly involved in an event, without implying effects on every connected company.
- **`subject_of`: Company → Event** - Identifies an event concerning a company. Inverse of `concerns`.
- **`reports`: SourceDocument → Event** - Connects a document to an event it describes.
- **`reported_in`: Event → SourceDocument** - Identifies documents describing an event. Inverse of `reports`.
- **`mentions`: SourceDocument → Company / Product / Service / Sector** - Records an explicit reference in a document. A mention alone does not establish a supply relationship.
- **`mentioned_in`: Company / Product / Service / Sector → SourceDocument** - Identifies documents referencing an entity. Inverse of `mentions`.

The inverse relationships define how each connection reads in reverse. Neo4j can traverse relationships in either direction, so these do not require duplicate stored edges.

Each extracted relationship retains supporting evidence, including the excerpt, source document, document type and filing or publication date. Where explicitly supported, supply relationships also identify the product or service supplied.

Question 2: Build it

I used **GPT-6 Luna through OpenRouter** to extract structured relationships from cleaned SEC document chunks and stored the resulting graph in **Neo4j**. Repeated assertions are consolidated into one relationship with multiple supporting excerpts.

The extraction prompt was drafted with GPT-6 Astra and included the ontology, permitted companies, positive examples and examples of what not to extract. Company names were masked consistently before extraction and restored before graph import to reduce reliance on prior company knowledge.

I also created a targeted, assistant-curated set of **20 labelled examples**, split into 12 training, four validation and four holdout examples, and ran **GEPA for three epochs**. No proposed revision outperformed the starting prompt on validation, so I retained it. The holdout was reused across experiments and should not be treated as a fresh unseen test.

A separate random spot-check of **50 extracted triplets** found:

- **47 supported by their sources (94%).**
- **Three incorrect or insufficiently supported (6%).**

This measures the quality of the sampled extracted triplets, not overall accuracy or recall: it does not count relationships the model failed to extract.

Broader spot-checks identified two recurring problems:

- **Invalid product entities.** The model sometimes treated vague wording as a defined product, for example `Lam Research → offers → limited product lines` and `Applied Materials → offers → new products`. These phrases appeared in the text but were not useful product entities.
- **Incorrect attribution.** Examples included `TSMC → offers → consumer GPUs` and `TSMC → offers → FPGAs`. Manufacturing products for customers does not establish that TSMC commercially offers those products itself.

These examples show why a supporting quote alone is insufficient: the quote must support the specific relationship and the intended meaning of its entities.

Question 3: Expose it

I built an MCP server with **nine tools** and connected it to an LLM through Open WebUI:

- **`get_company_basket`** - Lists the ten companies and their tickers.
- **`get_competitors`** - Finds a company's disclosed competitors, with supporting evidence.
- **`find_product_suppliers`** - Finds basket companies offering a specified product.
- **`trace_supply_chain`** - Traces suppliers or customers up to two connections away, with evidence.
- **`get_earnings`** - Retrieves historical earnings dates, actual EPS, consensus EPS estimates and beat/miss results.
- **`get_stock_chart`** - Displays recent stock prices and compares up to five companies on one chart.
- **`get_stock_history`** - Retrieves historical adjusted prices and compares returns over specified dates.
- **`find_company_documents`** - Searches archived SEC filings and exhibits by company, filing date and optional phrase.
- **`read_company_document`** - Reads an archived filing or earnings release, with citations linking to SEC excerpts.

Example question

> Assume ASML's production of new lithography equipment is halted for three months. Within our ten-company basket, identify first- and second-order downstream exposure. Show each supply path and its supporting evidence. Identify potentially exposed products where the documents support that connection, and distinguish documented dependencies from inferred impacts.

Example response, edited for brevity

The graph identifies **Intel and TSMC as directly connected customers**, with further potential exposure to **AMD, Broadcom, Intel and NVIDIA through TSMC**. "First-order" and "second-order" describe distance in the graph, not the severity or timing of an impact.

First-order exposure

- **ASML → Intel.** Intel disclosed that ASML was its sole supplier of EUV lithography tools deployed in Intel 4, Intel 3, Intel 18A and planned future leading-edge process nodes. These processes could face constraints on additional equipment, but the disclosure does not establish that existing production would stop. Intel 10-K, filed January 23, 2026
- **ASML → TSMC.** ASML identified TSMC as a key customer and described ASML tools operating in its chip-production facilities. This establishes an equipment relationship, but the cited passage does not identify particular tools or process nodes, quantify reliance or establish sole-supplier status. ASML annual-report exhibit, filed March 5, 2025

Second-order exposure through TSMC

- **ASML → TSMC → AMD.** AMD disclosed reliance on TSMC for all wafers used in its microprocessor and GPU products at 7 nm or smaller nodes. Those product families represent potential downstream exposure. AMD 10-Q, filed November 4, 2026
- **ASML → TSMC → Broadcom.** Broadcom reported that TSMC produced approximately 95% of the wafers manufactured by its contract manufacturers during the three fiscal quarters ended August 2, 2026. This indicates substantial foundry concentration, although the passage does not identify specific product lines. Broadcom 10-Q, filed September 10, 2026
- **ASML → TSMC → Intel.** Intel disclosed that some advanced current and future products were, or would be, manufactured exclusively by TSMC or depend on critical components made there, including compute dies. This creates an indirect exposure alongside Intel's direct dependence on ASML tools. Intel 10-K, filed January 23, 2026
- **ASML → TSMC → NVIDIA.** NVIDIA identified TSMC among the foundries producing its semiconductor wafers. Products using those wafers represent potential exposure, but the cited passage does not specify product lines, process nodes or TSMC's share of production. NVIDIA 10-K, filed February 25, 2026

These disclosures establish commercial connections and some product dependencies. They do **not** establish that a three-month halt in new ASML equipment production would stop installed tools from operating, exhaust equipment inventories or reduce wafer deliveries. The paths identify potential exposure requiring further investigation.

Question 4: Test one idea

I examined **six EPS-miss events over three years**, comparing each company with its graph-connected competitors over identical return windows. Competitors were grouped by their results for comparable fiscal quarters, even when they announced later. This makes the analysis retrospective rather than a trading strategy using only information available at the time.

Competitors that beat EPS expectations outperformed the company that missed in:

- **One trading day:** 16/21 comparisons (76%).
- **Five trading days:** 14/21 (67%).
- **Ten trading days:** 15/21 (71%).
- **One calendar month:** 9/21 (43%).

The one-day window suggested a potentially useful immediate reaction, but the results did not establish a consistent pattern across horizons. The comparisons also share companies and events, so they are not independent observations.

Revenue showed a stronger descriptive association: across **seven revenue-miss events**, revenue-beating competitors outperformed in **17/20 one-month comparisons (85%)**. However, removing Intel's two events changed the average peer-basket advantage from **+7.7 to −0.9 percentage points**. This sensitivity, alongside differences between consensus providers, limits the conclusion.

Overall, the results are **inconclusive**, with promising avenues for further research. Establishing a reliable pattern would require a 5-10-year backtest, a larger company basket, consistent historical consensus data, and sector and broad-market benchmarks. I would also examine earnings guidance, major news and deals during each return window: an EPS miss accompanied by strong forward guidance can still coincide with positive returns.

I would formally test whether the observed differences exceed what could arise by chance, using permutation tests that reassign beat/miss labels within comparable sectors and periods, and placebo tests using matched dates without earnings announcements. Confidence intervals should account for repeated companies and overlapping windows, with adjustments for testing multiple hypotheses and validation on an unseen period.

EPS and revenue surprises are important inputs, but their usefulness depends on context. The ontology helps connect financial results, company relationships and relevant events; statistical testing is needed to establish whether the observed associations are repeatable.